

Repetition Aversion and Revealed Preference

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- ▷ Ariel can rank restaurants in his neighborhood from best to worst. Why doesn't he always eat at the top restaurant?
- ▷ Bibi thinks that War and Peace is the best book ever written. Why doesn't he read it every month?
- ▷ Some “best things” are not necessarily always chosen, or even chosen more often
- ▷ But “ x is always chosen over y when available” (or “ x is chosen more often than y ”) is the standard empirical counterpart of preference for x over y
- ▷ How would Ariel and Bibi “reveal” their preferences over restaurants and books? What does their behaviour look like?

- ▷ We model choice with repetition aversion
- ▷ Treat it like a basic psychological (or physiological) trait, like delay aversion or risk aversion
- ▷ Distinct from (and generally not correlated with) fundamental preferences (= answers to: “what is the best...”)

An example: Ariel's routine

- ▷ Three restaurants; Ariel prefers $x \succ y \succ z$
- ▷ Ariel doesn't repeat choice in a restaurant till the menu has changed
- ▷ x changes menu every 2 weeks, y changes menu every 3 weeks, z changes menu every 4 weeks. We can interpret these as measures of Ariel's repetition aversion to restaurants

Weeks	1	2	3	4	5	6	7	8	...
Restaurant	x	y	x	z	x	y	x	z	...

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- ▷ Yet, z is chosen exactly as often as y , namely $1/4$ of the time
- ▷ Inferiority in both dimensions (preference and repetition aversiveness) is not revealed by (frequency of) choice.
- ▷ This is highly unusual in models of consideration sets

An example (ctd.)

- ▷ One day, x decides to change its menu less often, from once every 2 weeks to once every 3 weeks

Weeks	1	2	3	4	5	6	7	8	9	10	11	12	...
Restaurant	x	y	z	x	y	Skip	x	y	z	x	y	Skip	...

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- ▷ Now z is chosen every 6 weeks — the owner is very surprised to see Ariel more rarely because he is more bored with competitor!
- ▷ Increase in the repetition aversiveness of one alternative may harm other alternatives

- ▷ *Static taste for variety*. Kaiser and Schwabe (2012)
- ▷ *Habit formation or addiction*. E.g. Pollack (1970); Becker and Murphy (1988); Rozen (2010)
- ▷ *Dynamic Random Utility*. E.g. Turansick (2024)
- ▷ *Dynamic models of “taste for variety”*. E.g. McAlister (1982); Gilboa and Schmeidler (1997); Baucells and Sarin (2010); He, Dyer and Butler (2013); Puig (2024); Ashkenazi-Golan, Karos and Lehrer (2024)
- ▷ *Experimental implementation of repeated choice*: Agranov and Ortoleva (2017)
- ▷ *Foundations of stochastic choice*. Lu and Saito (2020)
- ▷ *Choice with consideration sets*. Masatlioglu Nakajima Ozbay (2012), Manzini and Mariotti (2014)

Model

- ▷ X is a finite set of alternatives and $a \notin X$ stands for abstention
- ▷ The (idealized) observable is a menu-choice sequence

$$(M, c) = (M_t, c_t)_{t=1}^{\infty}$$

taking values in $2^X \times \tilde{X}$, where $\tilde{X} = X \cup \{a\}$, such that $c_t \in M_t \cup \{a\}$

- ▷ Special case: $M_t = X$ for all t

Model: representation

- ▷ A linear order $\succ$ on $\tilde{X} = X \cup \{a\}$ such that $x \succ a$ for all $x \in X$
- ▷ A vector of aversion phases $p : \tilde{X} \rightarrow \{0, 1, 2, \dots\} \cup \{\infty\}$, $p(a) = 0$
- ▷ A vector of initial conditions $w_1 : \tilde{X} \rightarrow \{0, 1, 2, \dots\} \cup \{\infty\}$
- ▷ A tuple $(\succ, p, w_1)$ represents (M, c) if

$$\{c_t\} = \max(\{x \in M_t \cup \{a\} : w_t(x) \geq p(x)\}, \succ)$$

$$\text{where, for } t > 1, \quad w_{t+1}(x) = \begin{cases} w_t + 1 & \text{if } c_t \neq x \\ 0 & \text{if } c_t = x \end{cases}$$

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- ▷ $w_t(x)$ counts the number of periods x was not chosen before period t . If $w_t(x) < p(x)$, the agent is averse to x and does not consider it at t
- ▷ The agent chooses the best among feasible alternatives that the agent is not averse to (i.e. that the agent considers)

- ▷ Given c , define $\underline{w}_1(x) = 0$, $\overline{w}_1(x) = \infty$, and

$$\underline{w}_{t+1}(x) = \begin{cases} \underline{w}_t(x) + 1 & \text{if } c_t \neq x \\ 0 & \text{if } c_t = x \end{cases} \quad \overline{w}_{t+1}(x) = \begin{cases} \overline{w}_t(x) + 1 & \text{if } c_t \neq x \\ 0 & \text{if } c_t = x \end{cases}$$

- ▷ Given (M, c) , define the revealed preference

$$x \succ^* y \iff \exists t, s \in \mathbb{N} : c_t = x, y \in M_t, c_s = y, \text{ and } \underline{w}_t(y) \geq \overline{w}_s(y)$$

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Theorem 1. (i) A menu-choice sequence (M, c) is repetition averse if and only if $\succ^*$ is acyclic and $a \succ^* x$ for no $x \in X$.

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Theorem 1. (i) A menu-choice sequence (M, c) is repetition averse if and only if $\succ^*$ is acyclic and $a \succ^* x$ for no $x \in X$. (ii) Moreover, in this case, if all $x \in X$ are chosen, there is w_1 such that $(\succ, p, w_1)$ represents (M, c) if and only if $\succ^* \subseteq \succ$, and $\underline{p}(x, \succ) \leq p(x) \leq \overline{p}(x)$, where

$$\underline{p}(x, \succ) = \max(\{0\} \cup \{\underline{w}_t(x) + 1 : x \in M_t \text{ and } x \succ c_t\})$$

$$\overline{p}(x) = \min \{\overline{w}_t(x) : x = c_t\}$$

Theorem 1 proof sketch “if” part 1/2

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- ▷ Second, we show that $\underline{p}(x, \succ) \leq \bar{p}(x)$. If not, there are periods s, t such that $x = c_t$, $x \succ c_s$, $x \in M_s$, and

$$\underline{w}_s(x) + 1 = \underline{p}(x, \succ) > \bar{p}(x) = \bar{w}_t(x) \implies \underline{w}_s(x) \geq \bar{w}_t(x) \implies c_s \succ^* x$$

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- ▷ Third, given $\succ$ and p such that $\underline{p}(x, \succ) \leq p(x) \leq \bar{p}(x)$, define

$$w_1(x) = \max\{0, p(x) - \min\{t : c_t = x\} + 1\}$$

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- ▷ Let d be the choice induced by the model $(\succ, p, w_1)$. We show by induction in $t = 1, 2, \dots$ that $d_t = c_t$. Let's show the induction step

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 - ✓ Let $x = c_s$ for $s < t$, and, without loss, $x \neq c_r$ for $s < r < t$. Then

$$w_t(x) = \bar{w}_t(x) \geq \bar{p}(x) \geq p(x)$$

- ✓ Otherwise, x is chosen for the first time in c at t , and then

$$w_1(x) \geq p(x) - t + 1 \implies w_t(x) = w_1(x) + t - 1 \geq p(x)$$

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✓ Otherwise, y is not chosen in c and d before period t . Then

$$p(y) \geq \underline{p}(y) \geq \underline{w}_t(y) + 1 = t - 1 + 1 = t$$

$$w_t(y) = w_1(y) + t - 1 = \max\{0, p(y) - \min\{r : c_r = y\} + 1\} + t - 1 < p(y)$$

Dependence on the initial conditions

- ▷ Let $X = \{x, y, z\}$, $x \succ y \succ z \succ a$, and the aversion phases are $p(x) = 3$, $p(y) = 3$, $p(z) = 1$
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- ▷ The agent is not averse to any item: $w_1(x) = w_1(y) = w_1(z) = \infty$
 - ✓ The choice is $c = (x, y, z, a, x, y, z, a, \dots)$
 - ✓ Each alternative and the outside option are chosen with frequency $1/4$

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- ▷ Alternatively, $w_1(x) > 3$, $w_1(y) = 1$, and $w_1(z)$ is arbitrary
 - ✓ The choice is $c = (x, z, y, z, x, z, y, z, \dots)$
 - ✓ x and y are chosen with frequency $1/4$, and z is chosen with frequency $1/2$

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- ▷ If always offer the full menu: $M_t = X$ for all t ,
 - ✓ The choice is $c = (x, y, z, a, x, y, z, a, \dots)$
 - ✓ Frequencies of choices are $q(x) = q(y) = q(z) = 1/4$
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- ▷ In the long-run, the choice is associated with an absorbing directed cycle of the Markov chain. In particular, the choice is eventually cyclic
- ▷ If there are multiple absorbing cycles, the long-run choice frequencies may depend on the initial aversion conditions and menus

Extensions

Extension: Repetition Aversion with Memory Horizon

- ▷ Think about $M_t = \{x\}$ and $c = (x, x, a, x, x, a, x, x, a, \dots)$
- ▷ A parsimonious way to accommodate such choice: “choose x in any period in which x has not been chosen in at least one of the last two periods”
- ▷ Another example $c = (x, x, a, a, x, a, a, x, a, a, a, x, a, a, x, a, a, x, a, a, \dots)$

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- ▷ Another example $c = (x, x, a, a, x, a, a, x, a, a, a, x, a, a, x, a, a, x, a, a, \dots)$
- ▷ Generalize a model to allow the repetition aversion (consideration) to depend on the recent choice history in a more complicated way
- ▷ To ease notations and analysis, make two additional assumptions:
 - ✓ Assume that the menu is fixed, $M_t = X$ for all t
 - ✓ Assume that there is no initial aversion; then the order of first choices of alternatives reveal the linear order $\succ$ (i.e. intrinsic preferences)
- ▷ Can relax these assumptions; closer to Theorem 1 (“revealed preferences”)

- ▷ Let a natural T denote the horizon of the choice history that matters for the agent. Give a choice sequence c , define

$$h_t^T(c) = \begin{cases} (c_{t-1}, c_{t-2}, \dots, c_{t-M}) & \text{if } t > M \\ (c_{t-1}, \dots, c_t, a, \dots, a) & \text{if } t \leq M \end{cases}$$

- ▷ Implicit here is the “no initial aversion” assumption

Partial histories and monotonicity

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- ▷ Want monotonicity: more often and more recent consumption of x makes the agent more averse to x

✓ Define $\delta : \tilde{X} \times \tilde{X}^T \rightarrow \{0, 1\}^T$ via $(\delta_x(h))_i = \mathbb{1}\{h_i = x\}$

✓ For $l, k \in \{0, 1\}^T$, define $l \supseteq k \iff \sum_{i=1}^m l_i \geq \sum_{i=1}^m k_i$ for all $m = 1, \dots, T$

✓ A function $f : \{0, 1\}^T \rightarrow \mathbb{R}$ is weakly decreasing with respect to $\supseteq$ if

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- ▷ Denote xEy if x is chosen for the first time before y

$$\{c_t\} = \max \left(\left\{ x \in \tilde{X} : R(x, \delta_x(h_t^T(c))) = 1 \right\}, \succ \right)$$

- ▷ $\succ$ is a linear order on $\tilde{X} = X \cup \{a\}$ such that $x \succ a$ for all $x \in X$
- ▷ $R : \tilde{X} \times \{0, 1\}^T \rightarrow \{0, 1\}$ is a consideration function such that:
 - ✓ $R(a, h) = 1$ for all h
 - ✓ $R(x, (0, \dots, 0)) = 1$ for all x
 - ✓ $R(x, \cdot)$ is weakly decreasing with respect to $\supseteq$ for all x

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Theorem 2. (i) c is T -repetition averse if and only if aEx for no $x \in X$ and

$$\forall s, t, \quad xEy, \ c_t = x, \ c_s = y \implies \delta_x(h_t^T(c)) \not\supseteq \delta_x(h_s^T(c))$$

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(ii) Moreover, in this case, if all $x \in X$ are chosen, $(\succ, R)$ represents c if and only if $\succ = E$ and $\underline{R}(x, h) \leq R(x, h) \leq \bar{R}(x, h)$, where

$$\underline{R}(x, h) = \mathbb{1} \left\{ \exists t \in \mathbb{N} : c_t = x \text{ and } h_t^T(c) \supseteq h \right\}$$

$$\bar{R}(x, h) = 1 - \mathbb{1} \left\{ \exists t \in \mathbb{N} : xEc_t \text{ and } h \supseteq h_t^T(c) \right\}$$

Theorem 2 proof sketch “if” part

- ▷ Focus on the case when all $x \in X$ are chosen. Let $\succ = E$ and

$$\underline{R}(x, h) \leq R(x, h) \leq \overline{R}(x, h)$$

- ▷ If $\underline{R}(x, h) > \overline{R}(x, h)$ for some x, h , there are s, t : $c_t = x$, xEc_s , and

$$\delta_x(h_t^T(c)) \supseteq \delta_x(h), \quad \delta_x(h) \supseteq \delta_x(h_s^T(c)) \implies \delta_x(h_t^T(c)) \supseteq \delta_x(h_s^T(c))$$

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- ▷ Let $x = c_t$, then x is considered by d at t :

$$R(x, \delta_x(h_t^T(d))) = R(x, \delta_x(h_t^T(c))) \geq \underline{R}(x, \delta_x(h_t^T(c))) = 1$$

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- ▷ Consider $y \succ c_t$, then yEc_t , and y is not considered by d at t :

$$R(y, \delta_y(h_t^T(d))) = R(y, \delta_y(h_t^T(c))) \leq \overline{R}(y, \delta_y(h_t^T(c))) = 0$$

Extension: monotonic repetition aversion

- ▷ Suppose gradual accumulation of past experiences makes repetition aversion more acute
- ▷ Think about $M_t = \{x\}$ and $c = (x, a, x, a, a, x, a, a, a, x, a, a, a, a, x, \dots)$
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- ▷ A model with a finite memory horizon cannot accommodate such choice: the choice is not eventually cyclic
- ▷ Generalize the baseline model to allow the repetition aversion phases to increase with time
- ▷ To ease notations and analysis, make two additional assumptions:
 - ✓ Assume that the menu is fixed, $M_t = X$ for all t
 - ✓ Assume that there is no initial aversion; then the order of first choices of alternatives reveal the linear order $\succ$ (i.e. intrinsic preferences)
- ▷ Can relax these assumptions; see the paper, Appendix B

$$\{c_t\} = \max \left(\left\{ x \in \tilde{X} : w_t(x) \geq p(x, n_t^c(x)) \right\}, \succ \right)$$

- ▷ $\succ$ is a linear order on $\tilde{X} = X \cup \{a\}$ such that $x \succ a$ for all $x \in X$
- ▷ w_t counts the number of periods x was not chosen before period t , with $w_1(x) = \infty$ (no initial aversion)
- ▷ $n_t^c(x)$ is the number of times x is chosen before period t :

$$n_t^c(x) = |\{s \in \{1, \dots, t-1\} : c_s = x\}|$$

- ▷ $p : \tilde{X} \times \{0, 1, 2, \dots\} \rightarrow \{0, 1, 2, \dots\} \cup \{\infty\}$ is an aversion phase,
 - ✓ $p(a, k) = 0$ for all k
 - ✓ $p(x, 0) = 0$ for all x
 - ✓ $p(x, \cdot)$ is non-decreasing for all x

- ▷ Denote xEy if x is chosen for the first time before y in c
- ▷ Note that $w_t(x)$ is determined by c

Theorem 3. (i) c is monotonic repetition averse if and only if aEx for no $x \in X$ and

$$\forall s, t \text{ such that } t > s, \quad c_t E c_s \implies w_t(x) > w_s(x)$$

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(ii) Behaviorally equivalent to the model in which the aversion phase p of an alternative is a non-decreasing function of time (iii) Identification results: see the paper

Long-run frequencies, stochasticity, discussion

- ▷ Even if the model is deterministic, it can be viewed as generating a stochastic choice function
- ▷ Assume we observe sequences of choices from multiple fixed menus, i.e. from subsets of X (with a). That is, $M_t = A$ for all t for different $A \subseteq X$
- ▷ Identify the long-run frequencies of choice with probabilities of choice
- ▷ Denote by $\varphi(x, \tilde{A})$ the frequency of $x \in \tilde{A} = A \cup \{a\}$ in menu A

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- ▷ Identify the long-run frequencies of choice with probabilities of choice
- ▷ Denote by $\varphi(x, \tilde{A})$ the frequency of $x \in \tilde{A} = A \cup \{a\}$ in menu A
- ▷ Recall **Regularity**: $\phi(x, \tilde{A}) \geq \phi(\tilde{A} \cup \{y\})$
- ▷ Recall **Luce Independence**: for $x, y \in \tilde{A} \cap \tilde{B}$,
$$\frac{\phi(x, \tilde{A})}{\phi(y, \tilde{A})} = \frac{\phi(x, \tilde{B})}{\phi(y, \tilde{B})}$$

Example: violation of Regularity (and Luce Independence)

- ▷ $X = \{x, y, z\}$, $x \succ y \succ z$
- ▷ $p(x) = 1$, $p(y) = 2$, $p(z) = 3$
- ▷ No initial aversion: $w_1(x) = w_1(y) = w_1(z) = \infty$

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- ▷ Let $M_t = A = \{y, z\}$
 - ✓ The choice is $c = (y, z, a, y, a, z, y, a, a, y, z, a, y, a, z, y, a, a, \dots)$
 - ✓ $\phi(z, \tilde{A}) = \frac{2}{9}$

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 - ✓ $\phi(z, \tilde{A}) = \frac{2}{9}$
- ▷ Let $M_t = B = \{x, y, z\}$
 - ✓ The choice is $c = (x, y, x, z, x, y, x, z, \dots)$
 - ✓ $\phi(z, \tilde{A} \cup \{x\}) = \phi(z, \tilde{B}) = \frac{1}{4} > \frac{2}{9} = \phi(z, \tilde{A})$

Example: violation of Luce Independence with Regularity holding

- ▷ $X = \{x, y, z\}$, $x \succ y \succ z$
- ▷ $p(x) = 3$, $p(y) = 4$, $p(z) = 5$
- ▷ No initial aversion: $w_1(x) = w_1(y) = w_1(z) = \infty$

Example: violation of Luce Independence with Regularity holding

- ▷ $X = \{x, y, z\}$, $x \succ y \succ z$
- ▷ $p(x) = 3$, $p(y) = 4$, $p(z) = 5$
- ▷ No initial aversion: $w_1(x) = w_1(y) = w_1(z) = \infty$
- ▷ Let $M_t = D = \{x, z\}$

$$c = (x, z, a, a, x, a, a, z, x, a, a, a, x, z, a, a, x, a, a, z, x, a, a, a, \dots)$$

$$\phi(x, \tilde{D}) = \frac{1}{4}, \quad \phi(z, \tilde{D}) = \frac{1}{6} = \frac{5}{30}$$

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$$\phi(x, \tilde{D}) = \frac{1}{4}, \quad \phi(z, \tilde{D}) = \frac{1}{6} = \frac{5}{30}$$

- ▷ Let $M_t = B = \{x, y, z\}$

$$c = (x, y, z, a, x, a, y, a, x, z, a, y, x, a, a, z, x, y, a, a, x, z, y, a, x, a, a, y, x, z, a, a, x, y, a, z, \dots)$$

$$\phi(x, \tilde{B}) = \frac{1}{4}, \quad \phi(z, \tilde{B}) = \frac{5}{32}$$

- ▷ Even in the baseline model, the aversion phases may be not fully identified when $M_t = X$ is fixed or cyclic: the choice data is not rich enough
- ▷ Imagine M_t is a realization of a stochastic process. Say, M_t is drawn i.i.d. from full-support distribution on subsets of X .
 - ✓ Can identify $p(x) = \min\{\#\text{periods between two choices of } x\}$
 - ✓ Can identify $\succ$
- ▷ When aversion phases are large enough, can actually identify $\succ$, p , and the distribution of menus even if the latter is unknown
- ▷ The key is ergodicity: observing a single sequence of choices allows the analyst to identify all conditional frequencies of choices

Deterministic model as a limit of the stochastic model

- ▷ Suppose each alternative is present in the menu M_t with probability $1 - \epsilon_x$, independently
- ▷ The baseline model then induces a stochastic choice
- ▷ As discussed, a single realization suffices to restore the distribution
- ▷ What happens when $\epsilon_x \rightarrow 0$ for all x ?

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- ▷ As discussed, a single realization suffices to restore the distribution
- ▷ What happens when $\epsilon_x \rightarrow 0$ for all x ?
- ▷ The menu remains $M_t = X$ for very long periods of time; the choice is associated with the absorbing cycles of the deterministic Markov chain
- ▷ Sometimes, the cycles can “mutate” into each other when some alternatives are absent from the menu
- ▷ Can calculate the resulting long-run frequencies as weighted averages of the frequencies associated with choice cycles

- ▷ A parsimonious “ordinal” model of repetition aversion, assuming that a single realization of the choices is observed
- ▷ Characterization: constructing the revealed preference and demanding the Strong Axioms of Revealed Preference
- ▷ Identification: partial for both the intrinsic preference and the aversion parameters. Stochasticity helps to get full identification in the baseline model, but not for the extensions
- ▷ Initial conditions can substantially influence long-run choice frequencies in some but not in all cases
- ▷ Multiple connections to the stochastic choice
- ▷ More models in the paper, and some not even in the paper